BASE CHANGE OF A STANDARD V COVERING IS A STANDARD V COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3695, 3701] and proof(s) [[3703, 3716]]; this spec covers the complete environment instead (statement 3695-3701, proof span(s) [(3703, 3716)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Lemma 0.1. *Let $\{T_j \to T\}_{j=1,\ldots,m}$ be a standard V covering. Let $T' \to T$ be a morphism of affine schemes. Then $\{T_j \times_T T' \to T'\}_{j=1,\ldots,m}$ is a standard V covering.*

Proof. Let $\mathrm{Spec}(V) \to T'$ be a morphism where V is a valuation ring. By assumption we can find an extension of valuation rings $V \subset W$, an i, and a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(W) & \longrightarrow & T_i \\ \downarrow & & \downarrow \\ \mathrm{Spec}(V) & \longrightarrow & T \end{array}$$

By the universal property of fibre products we obtain a morphism $\mathrm{Spec}(W) \to T' \times_T T_i$ as desired. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.